EXAMINATIONS COUNCIL OF ESWATINI
Eswatini General Certificate of Secondary Education

BIOLOGY **6884/02**

Paper 2 **October/November 2027-2029**

Specimen **Total Marks: 80**

Confidential

MARK SCHEME

This document consists of **8** printed pages.

INTRODUCTION TO MARK SCHEME

General marking principles

1. Mark crossed out answers when nothing else has been written or if there is no second attempt at the answer. Otherwise ignore it.

2. If the candidate has left an answer blank, record this as NR (no response) in the marks column.

3. Apply rules consistently, e.g. in situations where candidates have not followed instructions.

4. Apply the agreed final mark scheme, making professional informed judgements about what the candidate has written in terms of context and correct use of scientific terms.

5. You need to decide if the point made is equivalent to the point in the mark scheme, or if it is not, based on your knowledge of the subject and understanding of the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind. Incorrectly used keywords should not be awarded marks.

6. Award marks in line with the standard of response required by a candidate as exemplified by the standardisation scripts.

7. Always award marks as whole marks (not half marks, or other fractions).

8. You should award marks using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

9. Only one mark should be awarded per numbered line, unless some of the lines are left blank, in which case the 2nd or 3rd idea on the first line can be marked as well.

10. Mark all the candidate's answers wherever they have written them.

11. Marks must be credited **positively**. This implies that you must:
    - award marks for correct/valid answers, as defined in the mark scheme. However, give credit for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader or Principal Examiner as appropriate
    - not deduct marks for errors and / omissions
    - judge answers only with respect to the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.
    - not deduct marks for spellings / grammatical constructions that don't matter (for many candidates, English is their 2nd or 3rd language)

Correct science marking principle

1. Always credit correct statements even if they follow incorrect statements. Usually apply this to sentences, but use judgement if candidate writes lengthy sentences.

2. Do not award mark for a correct statement if there is a contradictory statement(s) within the same part question, credit should not be awarded for any correct statement that is contradicted within the same question part. Ignore any incorrect science that is not related to the question.
3. You should consider the context and scientific use of any keywords when awarding marks (don't just spot key words / treat separate sentences independently). Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

4. Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection, ureter/urethra, iron/ion).

5. If first answer using information provided is incorrect, then allow ecf / TE for subsequent answers in the next question provided it is used correctly in the scientific context. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted. Additional guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

6. Do not allow 'particles' in place of molecules, ions, atoms.

Marking questions where a specified number of responses is indicated

For questions that require *n* responses (e.g. State **two** reasons …):

1. mark first answer on each row unless considered neutral.

2. if several answers on first line and no answers on subsequent lines, mark all answers on first line up to the number specified in the question.

3. do not mark answers in excess of number indicated by the question.

4. response should be read as continuous prose, even when numbered answer spaces are provided.

5. any response marked ignore in the mark scheme should not count towards *n*.

6. incorrect responses should not be awarded credit but will still count towards *n* (e.g. if 3 points are needed and 5 given, mark the first 3 only).

7. read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.

8. non-contradictory responses after the first *n* responses may be ignored even if they include incorrect science.

Guidance for calculations

1. Award full marks for correct answer and correct working with units, **unless units are given in question**.

2. If units are not given, then award a mark for numerical answer. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

3. If no answer or incorrect answer award marks for correct working and correct units.

4. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

5. For answers given in standard form (e.g. a × 10^n) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Guidance for chemical equations

1. Award full marks for a correct equation without state symbols unless a separate mark is awarded for the state symbols in the mark scheme or required in the question.

2. Accept multiples / fractions of coefficients used in chemical equations unless stated otherwise in the mark scheme.

Annotations

Use the system of annotations listed below as a shorthand for communicating marking decisions with Team leader or Principal Examiner.

Annotation	meaning
✓	correct point or mark awarded
✗	incorrect point or mark not awarded
^	information missing or insufficient for credit
A	allow or accept
I	incorrect or insufficient point ignored while marking the rest of the response
CON	contradiction in response, mark not awarded
BOD	benefit of the doubt given
ECF	error carried forward applied
NBOD	benefit of doubt was considered, but the response was decided to not be sufficiently close for benefit of doubt to be applied.
SEEN	point has been noted, but no credit has been given key point attempted / working towards marking point / incomplete answer / response seen but not credited
✓1	correct awarding one mark from marking point or marking group 1. similar numbered ticks are used for marking point or marking groups 2, 3, 4 etc.
⬭	used to highlight part of the response
PAG	point already given
MR	maximum mark reached
MAX	maximum number of marks for a marking point has been awarded.

Mark Scheme Abbreviations

Mark Schemes will use these abbreviations:

- ; separates marking points
- / separates alternatives for a marking point
- R reject
- A accept (for answers correctly cued by the question) or allow
- AW alternative wording (where responses vary more than usual)
- MP mark point- used in guidance notes when referring to numbered marking points
- ORA or reverse argument/reasoning
- AVP any valid point
- OWTTE or words to that effect
- I ignore/irrelevant – this response gains no mark, but any following correct answers can gain marks
- ( ) the word/ phrase in brackets is not required to gain marks but sets context of response for credit. e.g. (strong) covalent bond. Strong not needed but if it was described as weak covalent bond then no mark.
- Small underlined words this word only (grammatical variants excepted)
- D, L, T, Q quality of drawing/ labelling/ table / writing as indicated by mark scheme
- MAX indicates the maximum number of marks that can be given
- ECF / TE error carried forward/ transfer error
- SOI seen or implied

Question	Answer	Marks	Guidance
1(a)	*any two from*: nucleus; cytoplasm; cell membrane;	2	
1(b)	1 ref. to presence of cilia; 2 that move/waft/, to and fro **and** remove excess mucus/ dirt/ bacteria;	2	**MP2 A** - *sweeps away bacteria*
1(c)	*any four from*: ref. water potential gradient/ higher concentration of water molecules in distilled water than in cell; water enters; by osmosis; turgid cell/ vacuole enlarges/ pushes against cytoplasm; cell wall prevents bursting;	4	
2(a)	breaking down of large/insoluble food molecules into small/ soluble molecules; ref to. enzymes;	2	
2(b)(i)	peristalsis; contraction and relaxation; of circular and longitudinal muscles;	3	
2(b)(ii)	absorption/ digestion;	1	
2(c)	ORS, to replacing lost salts and fluids	1	
2(d)	**pancreas**- secrete insulin; **liver**- converts excess glucose + to glycogen; stimulates uptake of glucose by cells;	3	
2(e)	*any one from*: exercise; balanced diet / low-sugar diet; ref. to reducing carbohydrate in diet; ref.to maintaining a healthy body mass;	1	
3(a)	active is long-lasting / passive short lived / lasts for a short time; results in the formation of memory cells / no memory cells formed;	3	

	antibodies and lymphocytes generated when new antigens enter the body/ body cannot detect new infections;		
3(b)	fibrinogen; changes to fibrin; ref. to platelets;	**3**	
3(c)	$8 \div 1000 = 0.008$; 0.008×150; 1.2 (mm);	**3**	image size = actual size × magnification
4(a)	nutrition; sensitivity;	**2**	Ignore - respiration, growth
4(b)	1 at 0, no light + no photosynthesis; 2 as light intensity increases the rate of photosynthesis increases; 3 rate of photosynthesis does not increase with increasing light intensity after 3 au; 4 there must be a <u>limiting factor</u>;	**4**	**MP2 R** – ref. to temperature
4(c)	burning coal increases temperature in greenhouse to higher than outside; burning coal gives off carbon dioxide so level may be higher than outside; high temperature/higher carbon dioxide results in increased rate of photosynthesis;	**3**	
4(d)	less respiration; less energy; less active transport; less absorption of ions;	**4**	**A** - ref to yellowing
5(a)(i)	draw an arrow from the left hand side of the page towards the plant;	**1**	
5(a)(ii)	1 auxins; 2 accumulate on shaded side; 3 promote/stimulates growth/ cell division on shaded side; 4 positive phototropism;	**4**	**MP1 I** – hormones **MP2** and **MP3 - ORA**
5(b)	rods sensitive to dim light; so object can be seen;	**4**	

	cones detecting colour are sensitive to bright light; colour cannot be seen;		
6(a)	label line with **E** to oviduct;	**1**	
6(b)	*any two from*: cushions baby/shock absorber; prevents collapse of umbilical cord; allows movement; helps/encourages development of lungs; maintains constant temperature; lubrication during delivery;	**2**	
6(c)	ref. to oestrogen; stimulates repair (of uterus lining/ endometrium); ref. to progesterone; maintains thickness of the uterus lining/ increase blood vessels;	**4**	
6(d)	**in birth control:** prevent sperm from reaching oviduct; **in HIV prevention**: ensures there is no mixing of fluids;	**2**	
7(a)(i)	meiosis;	**1**	
7(a)(ii)	16;	**1**	
7(b)	parental genotypes $I^A I^o$ $I^B I^o$; gametes I^A I^o I^B I^o; offspring genotypes I^AI^B I^AI^o I^BI^o I^oI^o; offspring phenotypes AB A B O; correct labels (parental genotypes, gametes, offspring genotypes and offspring phenotypes ;	**5**	
7(c)	(chromosome) mutation; chromosome 21 fails to separate/segregate at meiosis/ gamate has 24 chromosomes; (zygote has) 47 chromosomes;	**2**	
7(d)(i)	have both normal and abnormal/ sickle-shaped, haemoglobin; red blood cells assume sickle cell shape; malaria parasite fails to enter sickle cells;	**3**	
7(d)(ii)	genetic variation;	**4**	

	ref. to changes in the environment; survival of fittest/ best adapted organisms survive; passing on the advantageous genes (to their offspring);		
8(a)	fungi; bacteria;	**2**	
8(b)	**G** nitrogen fixation; **H** decay/decomposition; **J** nitrification;	**3**	